

Repair Request - Performance Test Specification

Load / Stress / Spike / Soak + Evidence & Regression

Document Control	Value
Document ID	RR-PERF-001
Version	1.0 - Technical Performance Test Candidate
Status	Ready for Performance Test Design Approval
Revision Date	12 September 2026
Supersedes	New technical document; replaces prior draft performance appendix pages
Revision Scope	Provide a repeatable performance test method without inventing missing Business Requirements.
Approval	Solution Architect / Performance Test Lead / Business Owner for PERF-DEC-01 - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีเฉพาะ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

1. Scope / Non-Goals

This is a **Technical Performance Test Specification**, not a new Business Requirement document. The only existing numeric baseline is Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site. Other areas use diagnostic evidence until an approved measurable target exists.

PERF-DEC-01 - Performance Measurement Statistic (Pending Approval).
Business Baseline ระบุ “Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site” แต่ไม่ได้ระบุว่า 3 seconds หมายถึง Average, Maximum, p95 หรือ p99. จึง **ห้ามถือ p95 เป็น Requirement ที่อนุมัติแล้ว**. Recommendation สำหรับการอนุมัติคือ p95 end-to-end API response time <= 3.0 seconds; ก่อนอนุมัติ ให้บันทึก Average/p50/p95/p99/throughput/error rate ครบและรายงานผลเป็น diagnostic evidence โดยยังไม่ประกาศ Pass/Fail ของ release gate จาก percentile ไດ percentile หนึ่ง.

PTP-01 - Repeatable Load Test Profile (Technical Test Design)

Item	Technical Test Design - not a new Business Requirement
Baseline concurrency	100 concurrent virtual users - existing baseline.
Dataset	100,000 Repair Request records per Site - existing baseline. Related WO/Visit/Audit/ Notification data must be representative enough to exercise joins and security scope.
Security scope	Use at least 2 tenants and 2 Sites in the test fixture where practical; every request is authenticated. Cross-scope negative traffic is measured separately.
Ramp / duration	Default proposal: 5-minute ramp-up, 30-minute steady-state, 5-minute ramp-down, plus one warm-up run. This may be adjusted in a versioned Test Plan and is not a business requirement.
Filters / sorting / paging	Use representative filters, common sort, first/middle/deeper pages and the same server-side tenant/site scope as production code.
Evidence	Build/commit, environment, DB migration, dataset seed/version, script version, Average/p50/p95/p99, throughput, unexpected error count, API/DB latency, CPU/ memory, DB waits/blocking/slow-query evidence and trace samples.

2. Test Types

Type	Purpose	Gate
Load	Validate existing baseline under repeatable PTP-01 profile	

Type	Purpose	Gate
		Release/UAT gate only after PERF-DEC-01 approves the measurement statistic
Stress	Find saturation point and safe failure/recovery behavior above baseline	Diagnostic; no invented capacity target
Spike	Observe short burst, pool/queue recovery and unexpected error behavior	Diagnostic
Soak / Endurance	Detect memory/resource leak, backlog growth and connection exhaustion	Diagnostic; duration is versioned in execution plan

3. Layer Evidence Matrix

Layer	Required Evidence
Angular	Production build; network waterfall for critical list/search; superseded request cancellation evidence; render timing sample
API	.NET route Average/p50/p95/p99, throughput, unexpected errors, correlation/trace
EF Core	Generated SQL samples; query count per request; focused projection/AsNoTracking evidence on read paths
SQL Server/Azure SQL	Actual plan or Query Store evidence for critical paths; waits/blocking; CPU/resource metrics; long/slow query evidence
Worker	lag/backlog, claimed/completed/failed/retry counts, restart recovery, duplicate prevention evidence
File/Storage	bytes/duration/memory behavior, scan backlog and authorized download evidence

4. Performance Test Cases

ID	Scenario	Evidence / Acceptance
PERF-001	Normal List/Search	Existing <=3s baseline; selected statistic pending PERF-DEC-01. Always capture Average/p50/p95/p99, throughput, errors and scope correctness.
PERF-002	Filter/sort/paging/deeper pages	Bounded input; no unbounded query; generated SQL + plan evidence for representative variations.
PERF-003	Report/Export diagnostic	Record rows/bytes/time/memory and scope/PII correctness; no invented latency gate.
PERF-004	Worker load/retry/backlog	No duplicate logical event; record worker lag/backlog/retry/claim behavior and DB contention.
PERF-005	File I/O	

ID	Scenario	Evidence / Acceptance
		Record duration/bytes/memory/scan backlog; keep <=10MB/private/CLEAN/security semantics.
PERF-006	Concurrency contention	No lost update/duplicate transition; stale conflict remains controlled; collect lock/wait evidence.

5. Run Validity / Regression Workflow

A recorded run is valid only when environment/data/script/build versions are captured, monitoring is active, authentication/scope is representative and no unrelated deployment occurs during steady state. Store each approved run as a baseline artifact. Compare later runs against the last approved baseline; exact regression percentage is not invented and may be approved only after stable history exists.

6. Exit Criteria

- All required scripts/data/environment/evidence collection are ready.
- No unresolved data/scope correctness issue exists.
- PERF-DEC-01 is approved before a release Pass/Fail is claimed for the 3-second baseline.
- Production alert thresholds/on-call/RTO/RPO remain separate Production Gate decisions.